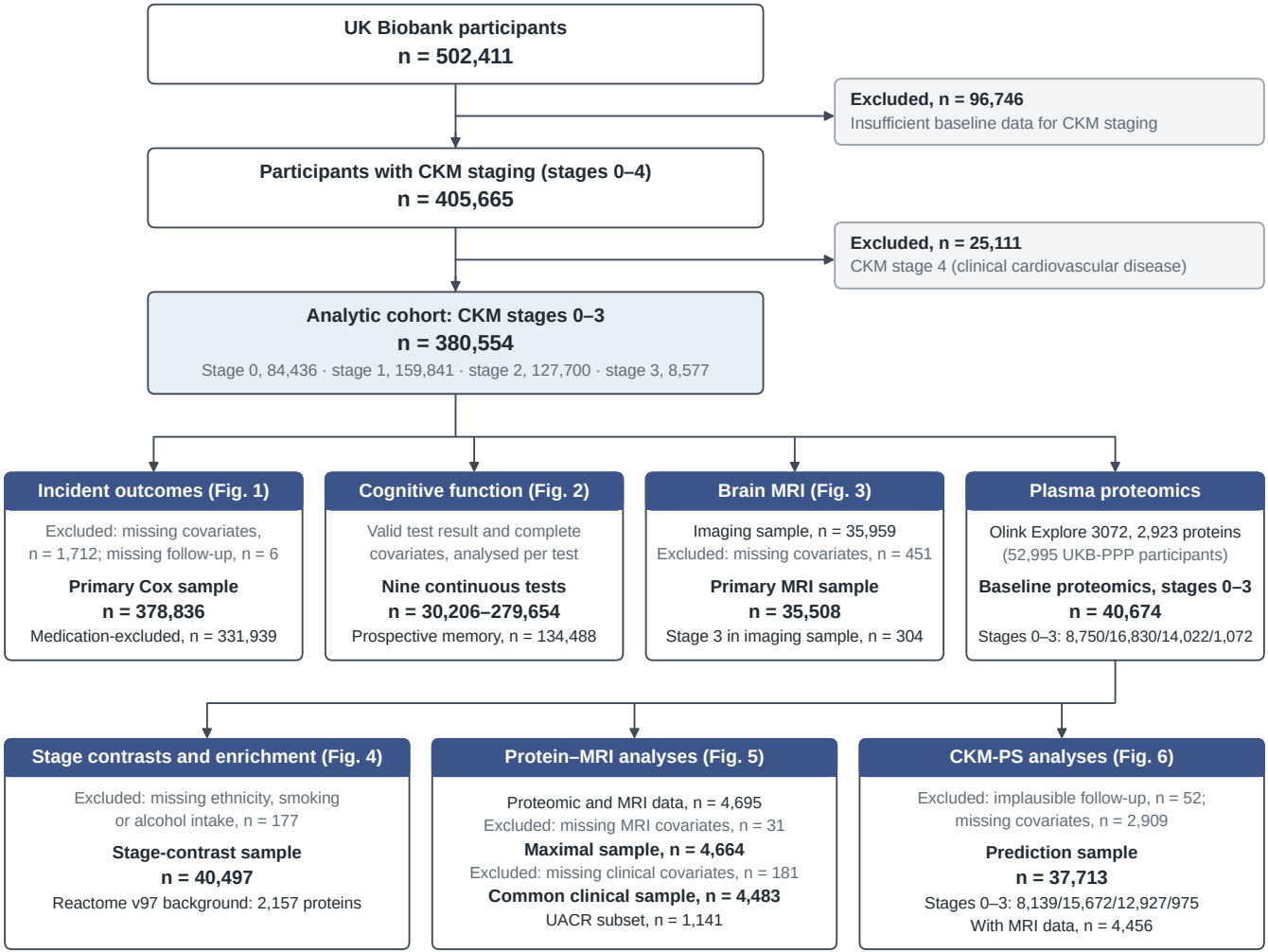
UK Biobank participants
n = 502,411
Excluded, n = 96,746
Insufficient baseline data for CKM staging
Participants with CKM staging (stages 0–4)
n = 405,665
Excluded, n = 25,111
CKM stage 4 (clinical cardiovascular disease)
Analytic cohort: CKM stages 0–3
n = 380,554
Stage 0, 84,436 · stage 1, 159,841 · stage 2, 127,700 · stage 3, 8,577
Incident outcomes (Fig. 1)
Excluded: missing covariates, n = 1,712; missing follow-up, n = 6
Primary Cox sample
n = 378,836
Medication-excluded, n = 331,939
Cognitive function (Fig. 2)
Valid test result and complete covariates, analysed per test
Nine continuous tests
n = 30,206–279,654
Prospective memory, n = 134,488
Brain MRI (Fig. 3)
Imaging sample, n = 35,959
Excluded: missing covariates, n = 451
Primary MRI sample
n = 35,508
Stage 3 in imaging sample, n = 304
Plasma proteomics
Olink Explore 3072, 2,923 proteins (52,995 UKB-PPP participants)
Baseline proteomics, stages 0–3
n = 40,674
Stages 0–3: 8,750/16,830/14,022/1,072
Stage contrasts and enrichment (Fig. 4)
Excluded: missing ethnicity, smoking or alcohol intake, n = 177
Stage-contrast sample
n = 40,497
Reactome v97 background: 2,157 proteins
Protein–MRI analyses (Fig. 5)
Proteomic and MRI data, n = 4,695
Excluded: missing MRI covariates, n = 31
Maximal sample, n = 4,664
Excluded: missing clinical covariates, n = 181
Common clinical sample, n = 4,483
UACR subset, n = 1,141
CKM-PS analyses (Fig. 6)
Excluded: implausible follow-up, n = 52; missing covariates, n = 2,909
Prediction sample
n = 37,713
Stages 0–3: 8,139/15,672/12,927/975
With MRI data, n = 4,456